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Subcortical and cerebellar volume differences in bilingual and monolingual children: An ABCD study

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ABSTRACT

Research suggests that bilingual children experience an extension or delay in the closing of the sensitive/critical period of language development due to multiple language exposure. Moreover, bilingual experience may impact the development of subcortical regions, although these conclusions are drawn from research with adults, as there is a scarcity of research during late childhood and early adolescence. The current study included 1215 bilingual and 5894 monolingual children from the ABCD Study to examine the relationship between subcortical volume and English vocabulary in heritage Spanish bilingual and English monolingual children, as well as volumetric differences between the language groups. We also examined the unique effects of language usage in bilingual children's subcortical volumes. In general, bilingual children had less cerebellar volume and greater volume in the putamen, thalamus, and globus pallidus than monolingual children. English vocabulary was positively related to volume in the cerebellum, thalamus, caudate, putamen, nucleus accumbens, and right pallidum in all children. Moreover, the positive relationship between vocabulary and volume in the nucleus accumbens was stronger for monolingual adolescents than bilingual adolescents. The results are somewhat in line with existing literature on the dynamic volume adaptation of subcortical brain regions due to bilingual development and experience. Future research is needed to further explore these regions longitudinally across development to examine structural changes in bilingual brains.

The study of bilingual language acquisition has gained great interest in recent research (De Houwer, 2017). Over half of the world's population is functionally bilingual (Grosjean, 2010; Wolck, 1988). Yet, monolingual language development is still considered the norm in various fields of research (i.e., monolingual bias; Barratt, 2018; Bhatia, 2017; De Houwer, 2023; Grosjean, 2008). In fact, understanding the development of the bilingual brain is an important topic in developmental neuroscience research (Byers-Heinlein et al., 2019; Claussenius-Kalman et al., 2021; Kroll and Fricke, 2014; Li, 2015; Petitto, 2009). Research has established that bilingual experience shapes the brain both functionally and structurally (Costa and Sebastián-Gallés, 2014; Li et al., 2014), and such neural adaptation can occur as early as infancy. While many studies have been dedicated to exploring the ways bilingual individuals learn their two languages (De Houwer, 2017), less research has examined the process by which children encode two languages in one brain. Thus, the current study aims to directly investigate these experience-dependent neuroanatomical profiles of monolingual and heritage bilingual children, especially subcortical gray matter volume.

Bi-and-multilingual children are exposed to multiple languages, some as early as birth. Bilingual infants are faced with complex sociolinguistic cues and must learn to efficiently process information to interact with the environment (Byers-Heinlein and Fennell, 2014). To develop appropriate mappings between the sound structure of the language to meaning or articulation, it is suggested that multiple language exposure takes advantage of developmental plasticity and that bilingual children may experience an extension or delay in the closing of the sensitive/critical period of language development (Berken et al., 2017).

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According to the sensorimotor hypothesis, areas associated with sensorimotor processing are implicated in bilingual language acquisition (Hernandez and Li, 2007). Specifically, neural regions responsible for sensorimotor learning and coordination, such as the basal ganglia, undergo rapid organization and reorganization early in development. As an early loss of plasticity can lead to difficulty in forming complex mappings later in life (Bates, 1999; Bates et al., 1997; Hensch, 2004; Pickett et al., 1998), the extension or delay of the closing of plasticity in bilingual individuals as described above may be an advantage for language learning specifically, and cognitive control more generally.

Prior research suggests that during early childhood, children undergo an increase in synaptic density and then a pruning process in later childhood and adolescence (Rapoport and Gogtay, 2007). Moreover, recent frameworks further emphasize the role of adolescence as a sensitive period of brain development (Fuhrmann et al., 2015). To date, only a few studies have directly compared the anatomical microstructure of bilingual and monolingual children (Pliatsikas et al., 2020; Vaughn et al., 2021). There is a lack of research on bilingual brain development during adolescence, as prior research primarily focused on infancy and adulthood (Costa and Sebastián-Gallés, 2014; Ramírez-Esparza and García-Sierra, 2014). Even when studies explore late childhood to early adolescence, they focus on the cortex rather than the subcortical areas (Vaughn et al., 2021; see Pliatsikas et al., 2020 for an extensive review). One study by Archila-Suerte and colleagues (2018) examined bilinguals between 6 and 13 years old and found that compared to unbalanced bilinguals, balanced bilinguals had larger putamen, a structure related to articulatory control (Pliatsikas et al., 2017) and procedural-based aspects of language (Ullman, 2020).

Subcortical regions have been implicated in second language (L2) processing and control in prior research (Green and Abutalebi, 2013; Li et al., 2014). The Adaptive Control Hypothesis (Green and Abutalebi, 2013) suggests that the basal ganglia, cerebellum, and thalamus may have different roles depending on the bilingual language environment. Specifically, the thalamus may be involved in language selection in bilinguals (Green and Abutalebi, 2013). Moreover, the basal ganglia nuclei (e.g., caudate, putamen, globus pallidus, nucleus accumbens) have been implicated in the modulation of cortical inputs to prefrontal regions. It is suggested that the frontostriatal pathways may be strengthened by the bilingual experience (Stocco et al., 2014). In addition, the cerebellum is associated with grammatical processing in bilinguals (Pliatsikas et al., 2014). Due to their contribution to second language processing, the current study focuses on the caudate, putamen, thalami, nucleus accumbens, globus pallidus, and cerebellum.

During development, some structures of the subcortical regions (i.e., globus pallidus, cerebellum) show an increase, peaking around age 15, followed by a decrease in gray matter volume, while other regions (i.e., caudate, putamen, nucleus accumbens) show a linear decrease in gray matter volume from childhood to adulthood (Herting et al., 2018; Sussman et al., 2016; Tiemeier et al., 2010; Wierenga et al., 2014). Based on research in adults, Pliatsikas and colleagues (2020) proposed that across development, bilingual individuals may show greater subcortical volume than monolingual individuals, but only after bilinguals have experienced a substantial period of bilingualism. In a study examining a cross-sectional sample of individuals from 3 to 21, they did not find differences between monolinguals and bilinguals in the subcortical regions or the cerebellum (Pliatsikas et al., 2020). This is somewhat consistent with our recent findings showing a lack of differences between bilingual and monolingual young adults. Although we found variability within groups in gray matter volume of the thalamus, nucleus accumbens, and globus pallidum associated with language proficiency and age of acquisition (Xu et al., in review).

It must be noted that Pliatsikas and colleagues (2020) examined a sample of bilinguals whose first language (English) is the same as the monolingual's first and only language (L1). In addition, they do not have information about the age of acquisition, which could be attributed to the null effect found across development. In bilingual research, L2 is often the weaker and the later-learned language; few studies have examined *heritage bilingual speakers*. In the United States, 13.2% of the population five years and over reported speaking Spanish in addition to English (Census, 2020). Children who grow up in Spanish-speaking homes and have some degree of proficiency in Spanish are considered *Spanish heritage speakers* (Valdés, 2014). They are usually enrolled in L2 education from an early age and have a more dominant L2 instead of L1. This intense L2 learning experience and pattern of flipped language dominance may lead to unique patterns of subcortical area adaptation in children, which can lead to possible differences between monolinguals and bilinguals not previously observed. Prior research in young adult heritage speakers does not reveal structural differences between the two groups, but variability within each group in terms of subcortical structures (Xu et al., in review). Other research shows cortical differences between groups in childhood but not in adulthood (Nguyen et al., 2023; Vaughn et al., 2021). It is unclear whether heritage speakers may show subcortical differences earlier in development but not in adulthood, similar to patterns observed in the cortical thickness of similar samples. The current paper explores the subcortical differences between English monolinguals and early heritage Spanish bilinguals using data from the Adolescent Brain Cognitive Development Study (ABCD; Casey et al., 2018; Garavan et al., 2018; Luciana et al., 2018)—a longitudinal study that collected neural and cognitive data from a large sample of American children throughout adolescence. In addition, we explored the relationship between subcortical volume and language usage in bilinguals. While prior findings showed a relationship between usage and cortical thickness in bilingual children (Vaughn et al., 2021), it is unclear whether there are individual differences in subcortical volume in relation to English use. Examining the relation between neuroanatomy and language experience will aid in the identification of neural correlates in emerging bilingual children and allow a better understanding of the development of the bilingual brain. We ask the following questions: 1) Are there subcortical volume differences between monolinguals and bilinguals? 2) Does the subcortical volume relate to English vocabulary and does this differ between monolingual and bilinguals? and 3) Does the subcortical volume relate to language usage in bilinguals?

1. Method

1.1. Participants

Participants in the current study consisted of children selected from the ABCD study. The ABCD study follows over 11 thousand children across 21 sites in the United States beginning at nine and ten years old. Data from the current paper are obtained from the ABCD data release 4.0 (doi: 10.15154/1523041), and analyses utilized the baseline structural MRI data. We first selected participants with complete, quality-controlled, and protocol-compliant T1-weighted, T2-weighted, and diffusion MRI data without clinical MRI findings (e.g., hydrocephalus, herniation). Then, children missing demographic data from the Parent Longitudinal Demographic Questionnaire (PLDQ; collected during the one-year follow-up visit) were excluded.

Using the answers from the PLDQ, we identified two language groups: monolinguals and bilinguals. Monolinguals were defined as children whose parents reported all the following: (1) child's native language was English, (2) English was used more than any other language at home, and (3) the child had never been enrolled in a dual language program at school. Bilinguals were children whose parents reported that either a) the child's native language was not English or b) the child's native language was English, but English is used equal to or less than another language in the home. Children whose *self-reported* language knowledge conflicted with their grouping (i.e., children in the "bilingual" group who reported that they did not speak a language other than English and children in the "monolingual" group who reported that they did speak a language other than English) were further removed from the sample. This resulted in a final sample of 1215 bilingual and

5894 monolingual children. Since the sample of monolinguals is much larger than that of bilinguals, all analyses include relevant demographic information that may differ between the two language groups, including age, biological sex, pubertal status, household income, parent education, nonverbal IQ, and handedness.

2. Measures

2.1. Questionnaires and behavioral tasks

Various measures were extracted from the ABCD study to be included as covariates in all analyses. Child's biological sex,age, parent education, and combined household income came from the Parent Longitudinal Demographic Questionnaire (modified from the PhenX toolkit; Hamilton et al., 2011). Handedness was measured using the Edinburgh Handedness Inventory (Oldfield, 1971; Veale, 2014). Pubertal status was measured using the Pubertal Development Scale (Barch et al., 2018; Petersen et al., 1988). Nonverbal IQ was measured using an automated version of the Matrix Reasoning subtest from the Wechsler Intelligence Test for Children-V (WISC-V) (Wechsler, 2014). For a more detailed summary of these measures, review the supplementary materials. See Luciana and colleagues (2018) for a complete description.

2.1.1. English picture vocabulary

English receptive picture vocabulary was measured using the NIH Toolbox Picture Vocabulary Task (Gershon et al., 2014) using an iPad. The task is adaptive to ensure appropriate difficulty for each child. Children see four pictures and are asked to touch the picture that matches the word they hear presented from an audio recording. For the current study, we used the age-corrected scores provided by the ABCD research team ($M = 100$; $SD = 15$). See Luciana and colleagues (2018) for more information. English language proficiency is a main predictor of interest.

2.1.2. Language use (Youth Acculturation Survey)

This measure was a subset of questions from the PhenX Acculturation protocol (Hamilton et al., 2011). Children were asked to rate their English ability (i.e., poor, fair, good, excellent). If children also reported speaking or understanding a language other than English, they were then asked to identify the other language. In addition, children rate their overall language use with family and their language use with friends on a 5-point scale (1 = Other language all of the time; 5 = English all of the time). See Zucker et al. (2018) for more information. Following Dick et al. (2019) approach to calculating "bilingual use" from this measure, we converted the 1–5 scale to a 0–4 scale where 0 = Other language all of the time and 4 = English all of the time. Then, we summed the ratings for language use with family and language use with friends to develop a 0–8 score, where 0 = Only other language with family & friends and 8 = Only English with family & friends. Scores in the middle of this range represent more balanced language use. A higher score represents higher English language use, as opposed to the other language. Bilingual language use is a main predictor of interest.

2.2. Neuroimaging

Neuroimaging data were collected at 21 different ABCD study sites on 3 T MRI scanners (Siemens Prisma, General Electric (GE) 750 and Philips) using standard adult-size head coils. MRI data were collected in a fixed order, which included 1) a 3D T1-weighted image; 2) a resting-state functional MRI (fMRI) scan; 3) diffusion tensor imaging (DTI); 4) a 3D T2-weighted image; 5) another resting-state fMRI scan; and 6) three task-based fMRI scans. Children watched a movie during the T1-weighted, T2-weighted and DTI scans. See Casey and colleagues (2018) for a complete description of the MRI scanning protocols and parameters.

2.2.1. Preprocessing

All data were analyzed using a collection of processing steps from the Multi-Modal Processing Stream at the University of California, San Diego (Hagler et al., 2019). Preprocessing involved distortion and motion correction and between-modality (i.e., T1-weighted; T2-weighted, DTI) registration. The details of these preprocessing steps are described elsewhere (Hagler et al., 2019). Data across scanners were harmonized using neuroCombat package in R (https://neuroconductor.org/help/neuroCombat/).

2.2.2. Subcortical and cerebellar volume

Subcortical volume (in mm^3) was extracted through FreeSurfer (Fischl, 2012). Each participant's structural MRI data were registered to an atlas in FreeSurfer, which was used to automatically label the subcortical structures. See Hagler and colleagues (2019) for a complete description. In the current study, the following bilateral ROIs were analyzed: the caudate, putamen, thalamus, nucleus accumbens, globus pallidus, and cerebellum.

2.3. Analyses

The secondary data analyses described in this paper were approved by Institutional Review Boards at the University of Texas Health Science Center at Houston and the University of Houston. Data for this study is available in the NIMH Data Archive, data release version 4.0 (doi: 10.15154/1523041).

2.3.1. Bilinguals and monolinguals subcortical volume correlates

As described above, monolingual and bilingual participants were selected from the ABCD study based on their parent- and self-reported experience with and knowledge of a language other than English. To compare the subcortical volume of the bilingual and monolingual groups with the selected ROIs mentioned above, we conducted a hierarchal regression analysis to explore the main and interaction effects of language background and English vocabulary, controlling for the covariates described above (i.e., age, sex, handedness, pubertal status, parent education, household income, and nonverbal IQ). English—the second language for the bilinguals and the first language for the monolinguals—is the societal language. The children in this study were born in the U.S. and have been enrolled in the U.S. school system. Thus, it would be meaningful to explore the association between subcortical volume and English proficiency as this is typically the stronger language for heritage speakers (Valdés, 2014).

Model 1 includes the unique effect of language group, model 2 includes the unique effect of language group and English proficiency, and model 3 includes the main effect of language group, the main effect of English proficiency, and the interaction between language group and English proficiency. After conducting the analyses for each region, we computed False Detection Rate (FDR) corrections (Benjamini and Hochberg, 1995) for all of the analyses for each outcome measure (volume of subcortical and cerebellar ROIs). Only results significant at an FDR-corrected alpha of 0.05 were reported.

2.3.2. Bilingual neural correlates of english use

We evaluated whether English use within bilinguals was related to subcortical and cerebellar volume. A multiple regression analysis was conducted, which included English use with friends and family as the main predictors while controlling for English vocabulary, age, sex, handedness, pubertal status, parent education, household income, and non-verbal IQ. Once again, after conducting these regressions for each region, we applied FDR corrections for all of the analyses for each outcome measure, and only reported the results that were significant at an FDR-corrected alpha of 0.05. This analysis allows us to explore the unique effect of English use in the bilingual adolescent brain.

3. Results

3.1. Demographic

See Table 1 for a comparison of demographic variables for bilingual and monolingual children. The two groups differed significantly in sex, race/ethnicity, handedness, and pubertal status, but not in age. Bilingual children had significantly lower household income and parent education, as well as nonverbal IQ than monolingual children. In addition, bilinguals had significantly lower English vocabulary than monolinguals, but the mean vocabulary scores of both bilingual and monolingual children were greater than the standardized mean of 100. Bilingual children, on average, reported balanced use of English and their other language with friends but more use of the other language with family. Bilingual children who reported using more English with family also reported using more English with friends ($r = 0.21$, $p < 0.001$). Bilingual children who reported using more English with both family and friends had higher English vocabulary scores ($r = 0.25$, $p < 0.001$).

See Table 2 for the relationship between English vocabulary and English use with the relevant covariates. Across the whole sample, English vocabulary positively correlated with age, nonverbal IQ, parent education, and household income. English vocabulary was negatively related to pubertal status, and males had higher vocabulary than females. English vocabulary was unrelated to handedness. Within bilinguals, English use was unrelated to age, sex, and pubertal status. English use was positively related to nonverbal IQ, household income, and parental education, and marginally differed by handedness.

Table 1
Demographic information for bilinguals and monolinguals.

	Bilinguals (n = 1,215)	Monolinguals (n = 5,894)	Test-Statistics for Comparison		
	n (%)		χ^2		***p***
Sex			$\chi^2(1) = .92$		.04
Male	630 (52%)	3,148 (53%)			
Female	585 (48%)	2,748 (47%)			
Ethnicity			$\chi^2(4) = 3434.1$		<.001
Asian	90 (7%)	50 (1%)			
Black	20 (2%)	911 (15%)			
Hispanic/ Latino	923 (76%)	435 (7%)			
White	122 (10%)	3832 (65%)			
Others	60 (5%)	665 (11%)			
Handedness			$\chi^2(2) = 11.73$		.003
Right	1,022 (84%)	4,725 (80%)			
Left	114 (9%)	750 (13%)			
Mixed	79 (7%)	419 (7%)			
Native Language					
English	362 (30%)	5,894 (100%)			
Spanish	679 (56%)		–		–
Other	174 (14%)				
	M (SD)		***t***	***95% CI***	***p***
Age in months	130.78 (7.54)	131.24 (7.76)	$t(1783.5) = -1.93$	[-0.03, 0.01]	.05
Pubertal Status	2.12 (0.84)	2.04 (0.83)	$t(5738) = 2.82$	[0.03, 0.14]	.004
Non-Verbal IQ	9.58 (2.98)	10.04 (2.93)	$t(6982) = -4.83$	[-0.64, -0.27]	<.001
Household Income	6.12 (2.54)	7.65 (2.24)	$t(1403.8) = -18.30$	[-1.69, -1.36]	<.001
Parent's Education	15 (4.02)	17 (2.27)	$t(1364.4) = -16.78$	[-2.24, -1.77]	<.001
English Vocabulary	100.67 (15.86)	108.65 (16.80)	$t(1820.5) = -15.81$	[-8.97, -6.99]	<.001
English Use					
With friends	4.25 (0.90)				
With family	2.39 (1.18)	–	–	–	–
Total	4.64 (1.62)				

Table 2
Relationship between English vocabulary and English use with covariates.

	English Vocabulary			Bilinguals' English Use[a]		
	Test-Statistic	*95% CI*	*p*	Test-Statistic	*95% CI*	*p*
Age	$r = .04$	[0.02, 0.06]	< .001	$r = .01$	[− 0.05, 0.07]	.74
Sex	$t(7107) = -3.82$	[− 2.32, − 0.75]	< .001	$t(1207) = -0.45$	[− 0.23, 0.14]	.65
Pubertal Status	$r = -.012$	[− 0.15, − 0.70]	< .001	$r = -.018$	[− 0.08, 0.05]	.58
Non-Verbal IQ	$r = .35$	[0.33, 0.37]	< .001	$r = .15$	[0.09, 0.20]	< .001
Household Income	$r = .37$	[0.35, 0.39]	< .001	$r = .31$	[0.25, 0.36]	< .001
Parental Education	$r = .37$	[0.35, 0.39]	< .001	$r = .25$	[0.20, 0.31]	< .001
Handedness	$F(2, 7106) = 0.87$		.418	$F(2, 1206) = 2.68$		.07

Notes. [a] Combined use with family and friends. r = Pearson's correlation. CI = Confidence Interval.

3.2. Subcortical volume differences between bilingual and monolingual children

See Table 3 for results. In model 1, when controlling for age, sex, handedness, pubertal status, household income, parent education, and nonverbal IQ, bilinguals had smaller cerebellums and larger putamen, globus pallidus, and right thalamus compared to monolinguals, although the effect sizes ($\eta_p^2 \sim <.001 - .037$) were small (Fig. 1; see supplemental Table 1 for full regression results including betas, standard error, FDR-corrected p-values, and effect sizes for all covariates). In model 2, when English vocabulary was included in the model, the same unique group differences were found, with the addition of larger left thalamus in bilinguals compared to monolinguals. Additionally, English vocabulary positively correlated with subcortical volume across the whole sample, such that greater English vocabulary was related to larger cerebellum, thalamus, caudate, putamen, nucleus accumbens, and right globus pallidus in both monolinguals and bilinguals, with small to medium effect size ($\eta_p^2 \sim .023 - .075$) (Fig. 2; see supplemental Table 2 for full regression results including betas, standard error, FDR-corrected p-values, and effect sizes for all covariates). Finally, in model 3, we explored the interaction between language groups and English vocabulary. In this model, the same vocabulary-subcortical volume correlation was found. In addition, bilinguals had larger nucleus accumbens compared to monolinguals. Interestingly, there was an interaction in these regions, such that the positive relationship between English vocabulary and nucleus accumbens volume was stronger for monolinguals than bilinguals, although the effect sizes ($\eta_p^2 = .003$) were small (Fig. 3; see supplemental Table 3 for full regression results including betas, standard error, FDR-corrected p-values, and effect sizes for all covariates).

3.3. Bilingual english use and subcortical volume

Within bilingual children, English use was unrelated to subcortical volume when controlling for English vocabulary, age, sex, pubertal status, nonverbal IQ, household income, parent education, and

Table 3
Subcortical volume correlates English vocabulary between bilinguals and monolinguals after adjusting for covariates.

Regions		Model 1			Model 2						Model 3								
		Group			Group			English Vocabulary			Group			English Vocabulary			Interaction		
		b	*SE*	η_p^2	*b*	*SE*	η_p^2	*b*	*SE*	η_p^2	*b*	*SE*	η_p^2	*b*	*SE*	η_p^2	*b*	*SE*	η_p^2
Left	Cerebellum	**-965.95** ***	173.94	0.037	**-900.54** ***	173.79	0.037	**22.67** ***	3.97	0.075	264.26	1110.32	0.037	**24.02** ***	4.17	0.075	-11.27	10.61	0.000
	Thalamus	40.61	22.37	0.008	**48.30** *	22.37	0.008	**2.67** ***	0.51	0.067	203.43	142.90	0.008	**2.84** ***	0.54	0.067	-1.50	1.37	0.000
	Caudate	6.44	16.11	0.007	11.99	16.11	0.007	**1.92** ***	0.37	0.046	88.42	102.90	0.007	**2.01** ***	0.39	0.046	-0.74	0.98	0.000
	Putamen	**77.67** ***	19.68	0.000	**81.72** ***	19.71	0.000	**1.40** **	0.45	0.029	114.12	125.91	0.000	**1.44** **	0.47	0.029	-0.31	1.20	0.000
	Globus pallidus	**25.61** **	8.42	0.001	**26.40** **	8.43	0.001	0.27	0.19	0.018	45.31	53.89	0.001	0.30	0.20	0.018	-0.18	0.52	0.000
	Nucleus accumbens	4.94	3.61	0.003	5.78	3.61	0.003	**0.29** ***	0.08	0.026	**71.48** *	23.06	0.003	**0.37** * **	0.09	0.026	**-0.64** *	0.22	0.002
Right	Cerebellum	**-943.93** ***	179.77	0.035	**-876.13** ***	179.62	0.035	**23.50** ***	4.11	0.075	141.15	1147.57	0.035	**24.67** * **	4.31	0.075	-9.85	10.97	0.000
	Thalamus	**56.35** **	19.83	0.005	**62.89** **	19.83	0.005	**2.27** ***	0.45	0.063	296.60	126.65	0.005	**2.54** * **	0.48	0.063	-2.26	1.21	0.001
	Caudate	-12.49	16.00	0.011	-6.84	16.00	0.011	**1.96** ***	0.37	0.048	79.50	102.19	0.011	**2.06** * **	0.38	0.048	-0.84	0.98	0.000
	Putamen	**104.77** ***	19.12	0.000	**108.63** ***	19.15	0.000	**1.34** **	0.44	0.032	117.02	122.36	0.000	**1.35** **	0.46	0.032	-0.08	1.17	0.000
	Globus pallidus	**29.20** ***	7.80	0.000	**30.61** ***	7.81	0.000	**0.49** **	0.18	0.024	43.25	49.92	0.000	**0.50** **	0.19	0.024	-0.12	0.48	0.000
	Nucleus accumbens	5.30	3.36	0.003	5.99	3.37	0.003	**0.24** **	0.08	0.023	**62.70** *	21.50	0.003	**0.31** * **	0.08	.024	**-0.55** *	0.21	0.001

Note. Bold indicate significant results. * $p < .05$, * * $p < .01$, * ** $p < .001$

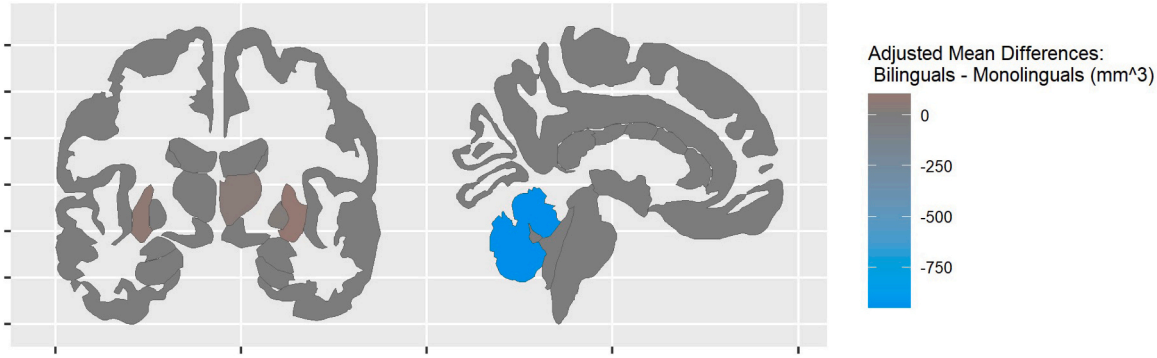

Fig. 1. Subcortical volume differences between bilinguals and monolinguals (model 1). Blue indicates less volume for bilinguals than monolinguals; Red indicates greater volume for bilinguals than monolinguals. All results presented were significant after controlling for age, sex, pubertal status, non-verbal IQ, household income, parent education, and handedness. Figure created using the ggseg R package (Mowinckel and Vidal-Piñeiro, 2020).

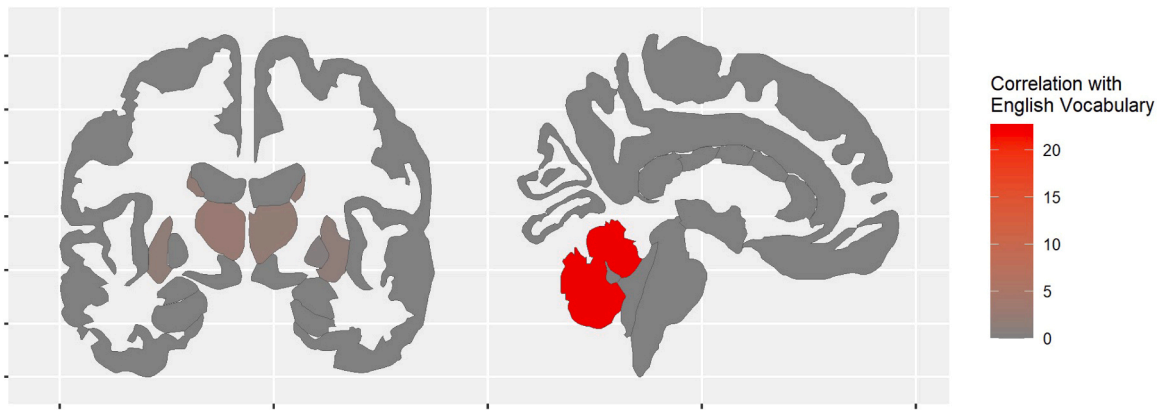

Fig. 2. Significant relationships between English vocabulary and subcortical volume (model 2). Red indicates greater volume associated with greater English vocabulary. All results presented were significant after controlling for age, sex, pubertal status, non-verbal IQ, household income, parent education, and handedness. Figure created using the ggseg R package (Mowinckel and Vidal-Piñeiro, 2020).

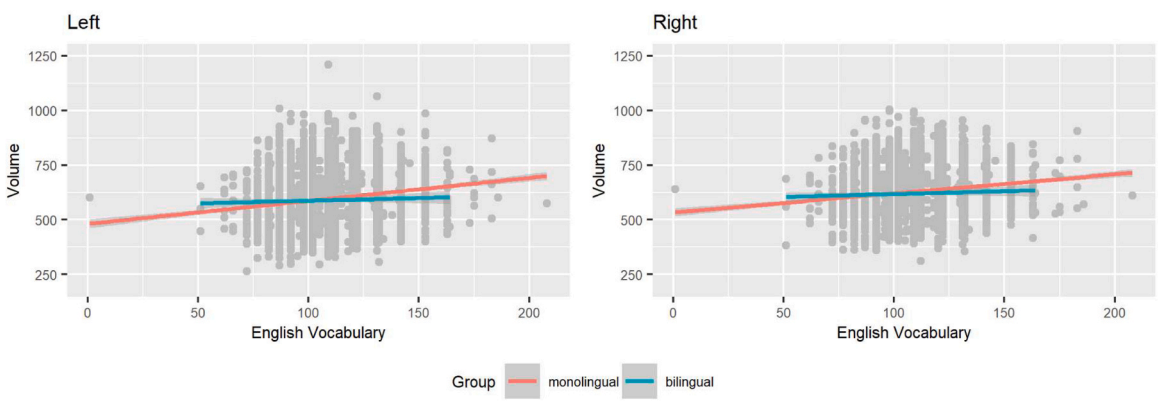

Fig. 3. Interaction between language group and English vocabulary at the nucleus accumbens (model 3). Left panel is left nucleus accumbens, right panel is right nucleus accumbens. All results presented were significant after controlling for age, sex, pubertal status, non-verbal IQ, household income, parent education, and handedness.

handedness.[5] See supplemental Table 4 for full regression results including betas, standard error, FDR-corrected p-values, and effect sizes for all covariates.

4. Discussion

The current study examined subcortical volume in monolingual and heritage bilingual adolescents. In this sample of 7109 children from the ABCD study, bilinguals had smaller cerebellum and larger putamen, thalamus, and globus pallidus than monolinguals. In all children, higher English vocabulary was related to larger bilateral cerebellum, thalamus, caudate, putamen, nucleus accumbens, and right globus pallidus. Notably, there was a language group by English vocabulary interaction

[5] Additional models were run including English usage but not English vocabulary in the model. Usage remained a non-significant predictor.

at the bilateral nucleus accumbens, such that the positive relationship between vocabulary and volume was stronger for monolingual than bilingual adolescents in these regions. Within bilinguals, English vocabulary and bilingual language usage were unrelated to subcortical volume.

4.1. Monolingual and bilingual differences in subcortical volume

Bilingual adolescents had larger putamen, thalamus, and globus pallidus than monolingual adolescents, even when controlling for English vocabulary. This finding is consistent with prior research on bilingualism and supports the previously implicated function of these regions in language control and processing. For instance, the putamen is implicated in articulatory control (Pliatsikas et al., 2017) and procedural-based aspects of language (Ullman, 2020). Immersed bilinguals showed greater putamen volume than monolingual adults (Abutalebi et al., 2013; Burgaleta et al., 2016). Similarly, balanced bilingual children had larger putamen than unbalanced bilingual children (Archila-Suerte et al., 2018). The thalamuson the other hand, is involved in language selection (Abutalebi and Green, 2016) and has differential activation between L1 and L2 in bilinguals (Sulpizio et al., 2020). Our findings are consistent with prior research showing greater thalamic volume in bilingual adults than in monolingual adults (Burgaleta et al., 2016). Finally, the globus pallidus' role has been suggested in the mediation of lexical–semantic processes (Crosson, 1985; Whelan et al., 2004), specifically in monitoring different stages of duo language speech production, from language selection to auditory-motor feedback during articulation (Liu et al., 2010).

Bilingual adolescents in the current sample also showed less cerebellum volume than monolingual adolescents. This result is inconsistent with prior research generally showing greater cerebellum volume with increased bilingual experience (Burgaleta et al., 2016; Deluca et al., 2019a,2019b; Filippi et al., 2011; Pliatsikas et al., 2014) as this region is implicated in bilingual syntactic processing (Ullman, 2020). A possible explanation of these findings and their opposite pattern compared to other regions above may concern the different patterns of development in the subcortical area. As previously mentioned, some structures of the subcortical regions show an increase, peaking around age 15, followed by a decrease in gray matter volume, while other regions show a linear decrease in gray matter volume from childhood to adulthood (Herting et al., 2018; Tiemeier et al., 2010; Wierenga et al., 2014). Moreover, multiple language exposure may lead to a delay in the closing of the sensitive/critical period of language development in bilingual children (Berken et al., 2017). During the age range of the current sample (9 to 10 years old), the cerebellum, which follows the non-linear increase then decrease in volume, would not yet have reached peak volume for either child group. It is possible that the delay of the critical period led to a delayed peak in volume, and thus bilingual children had smaller cerebellum than their monolingual counterparts at ages 9–10 (Fig. 4A). On the other hand, areas in the basal ganglia (i.e., putamen, globus pallidus) and the thalamus may follow the linear decrease in volume trajectory, and thus bilinguals had greater volumes than monolinguals during this stage of development (Fig. 4B).

It must be noted that a recent large study that compared monolinguals and bilinguals cross-sectionally between the ages of 3 and 21 did not find differences in subcortical volumes between groups (Pliatsikas et al., 2020). Our sample may differ from this sample of bilinguals in various important ways. For instance, in this prior work, the age of L2 acquisition was unclear. Thus, the lack of significant differences between monolinguals and bilinguals may be due to the variability in the bilingual language experience. In addition, this sample examined bilinguals whose first language is the same as the monolingual and is the societal language. In the current paper, bilingual children are heritage speakers whose home language (Spanish) differs from the societal language (English). These children are exposed to L2 early in life, usually before or by the time they enter the school system. Thus, our results may reflect the delaying of the closing of plasticity in heritage speakers as their brains must adapt to the complex linguistic environment. As there is a scarcity of developmental neuroscience bilingual research, future studies should continue to explore and extend these findings.

4.2. Language experience and subcortical volume

In general, English vocabulary was positively correlated with subcortical volume in every region except for the left globus pallidus. These findings are consistent with prior research showing a positive relationship between language experience and subcortical volume in bilingual adults in the putamen, thalamus, nucleus accumbens, and cerebellum (Marin-Marin et al., 2022; Xu et al., in review). Results are somewhat comparable to the trajectory proposed by the Dynamic Restructuring Model (DRM; Pliatsikas, 2020), which characterizes the general increase of subcortical volume with increasing second language experience. Combined with the sensorimotor hypothesis which emphasizes the development of the basal ganglia in L2 acquisition

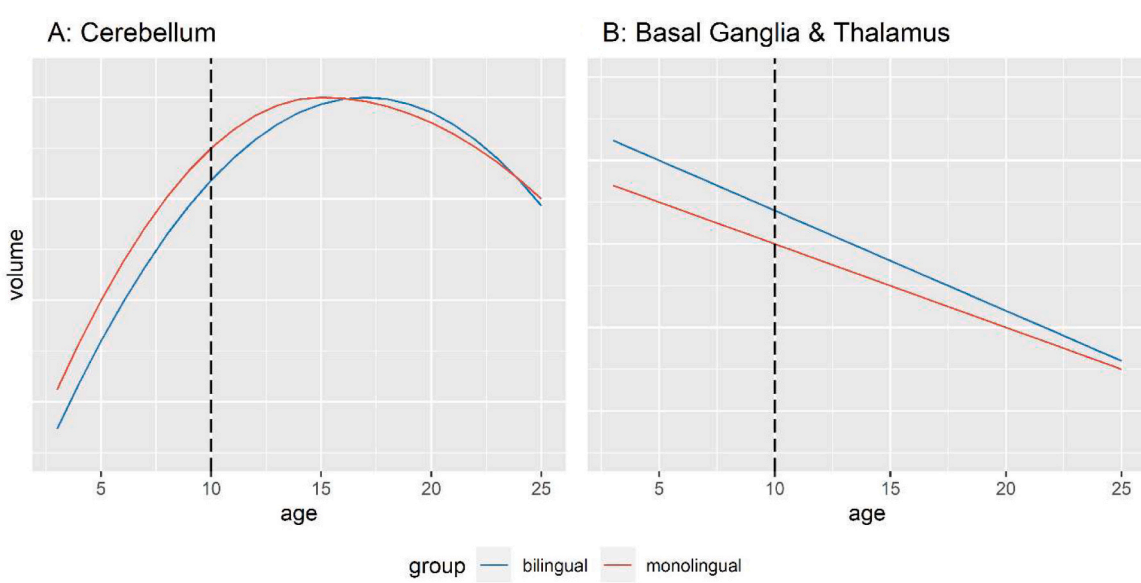

Fig. 4. A possible theoretical account for subcortical volumetric changes from childhood to adulthood in monolinguals and early bilinguals.

(Hernandez and Li, 2007), the current findings present evidence suggesting the adapting nature of these subcortical regions and their potential involvement in language processing in all children during development.

There was an interesting finding regarding the language group by English vocabulary interaction effect at the bilateral nucleus accumbens. Specifically, the positive relationship between English vocabulary and accumbens volume was stronger for monolingual than bilingual children. Consistent with the sensorimotor hypothesis (Hernandez and Li, 2007), the nucleus accumbens may be differentially involved in L2 compared to L1 acquisition, although our data cannot fully support this premise due to the lack of L1 vocabulary information for heritage bilinguals. Alternatively, the nucleus accumbens may be differently involved with language proficiency for bilingual compared to monolingual children due to differences in language control demands or exposure to English. Consistent with this explanation, prior findings in young adults also demonstrate a relationship between English proficiency and nucleus accumbens volume in heritage Spanish bilinguals but not in monolinguals (Xu et al., in review). The nucleus accumbens has been implicated in active language selection and processing (Deluca et al., 2019a,2019b). These structural findings are somewhat consistent with functional findings showing differential activation across language groups in a lexico-semantic task, such that monolinguals activated a broader set of cortical regions implicated in language control while bilinguals recruited a more localized subcortical network (Bradley et al., 2013). Thus, it is important for future research to further explore the role of this region in bilingual language acquisition and neural development. It must be noted that in the last analysis, which includes both main effects and an interaction term, there are no longer group differences. However, while the nucleus accumbens did not show group differences in the first two models, monolingual and bilingual children showed differences in nucleus accumbens volumes in the model which includes the interaction term. It is unclear why these differences between models emerge. It is possible that the differences between groups are highly dependent on English vocabulary, and monolinguals on average had higher English vocabulary than bilinguals. However, this does not explain the opposite pattern found between the cerebellum and other regions. Thus, future research is needed to further examine these findings.

Finally, within bilingual adolescents subcortical volume was unrelated to language usage. Previous findings in children also drawn from the ABCD sample showed a positive relationship between usage and cortical thickness in heritage bilingual children (Vaughn et al., 2021). The lack of usage unique effects in the current study is somewhat consistent with predictions that second language experience is reflected in the subcortical volume only after bilinguals have experienced a substantial period of bilingualism (Pliatsikas et al., 2020). More research is needed to explore these individual differences longitudinally in the bilingual brain.

4.3. Implications, limitations, and future directions

The current study brings novel research insight into examining language and subcortical volume with a large sample to explore the developmental aspect of language experience. We focused specifically on a previously less studied but prominent group–heritage bilingual children–in the language context of the U.S. where individuals learn L2 as the societal language and continue speaking a home language (as reflected in home and friend usage). The results provided support for prior research and framework concerning bilingual language development and offered additional explanations for neural patterns in an understudied group of bilinguals.

While the current study has many strengths, some conclusions cannot be made about heritage bilingual neural development and its relation to language experience due to the lack of L1 measures. In addition, while the ABCD study is immensely useful due to its representative and large sample, language measures already included in the study were not designed to examine other aspects of bilingual experience and thus cannot fully capture the potential effects of an immersive language experience. For example, the usage measure is a continuum from other language use to English use, rather than two separate questions, which may not fully capture the bilingual language environment. While having heritage bilinguals as our sample is a strength for diversity in bilingual research, the results may not generalize beyond other bilingual samples. In general, future research in bilingualism will benefit from a developmental cognitive neuroscience perspective. Thus, future longitudinal studies following bilinguals from childhood to adulthood can further the understanding of bilingual brain development. Specifically, future studies can take advantage of upcoming ABCD releases to explore the interaction between bilingualism and age. Given more time points, it is possible to model a nonlinear change in regions such as the cerebellum across development and whether this differs between bilingual and monolingual children.

Another aspect to mention is the small effects reported in the current study. The strength, and perhaps also weakness, of a large dataset such as the ABCD is the power to detect smaller effects at the critical value of $\alpha = .05$ (Dick et al., 2021). While the differences found in the current study between the monolingual and bilingual children are small, these differences in early adolescence (or lack of differences later in adulthood as found in Nguyen et al., 2023 and Xu et al., in review) may be meaningful in characterizing brain development across the lifespan in relation to single or multiple language experience. Indeed, our work and that of others highlight the large inter-individual variability in brain development. Little is known about individual variation in subcortical changes and should also be the focus of future research.

5. Conclusion

The current study examines subcortical volume differences between heritage Spanish bilingual and English monolingual adolescents, as well as the relationship between subcortical volume and English experience. The study extends bilingual developmental neuroscience research, supporting a pattern of extended neuroplasticity in bilingual children. Results are consistent with the sensorimotor hypothesis (Hernandez and Li, 2007) and further suggest the involvement of various subcortical areas in language acquisition and processing. In sum, bilinguals had smaller cerebellum and larger putamen, thalamus, and globus pallidus. In addition, higher English vocabulary is related to greater volume in the cerebellum, caudate, putamen, nucleus accumbens, and right globus pallidus. There was also a stronger relationship between vocabulary and volume at the nucleus accumbens in monolinguals than inbilinguals, suggesting potentially different involvement of this region in L1 and L2 language acquisition and processing. Bilinguals in this age range did not show volumetric variability in relation to bilingual language usage. These results highlight the neural difference between bilingual and monolingual adolescents and the dynamic nature of subcortical adaptation throughout bilingual language development. Longitudinal future research is needed to understand how the bilingual brain changes across development from childhood to adulthood.

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Code availability statement

Analysis scripts for this study are available on the Open Science Framework (doi: 10.17605/OSF.IO/UP7V8).

CRediT authorship contribution statement

Nguyen My Viet Ha: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Visualization. **Xu Yinan:** Validation, Writing – original draft, Writing – review & editing. **Vaughn Kelly A.:** Supervision, Validation, Writing – review & editing. **Hernandez Arturo E.:** Funding acquisition, Resources, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors do not have permission to share data. Data included in the secondary analyzes described in this paper was de-identified and obtained from the NIMH Data Archive under ABCD study data release version 4.0 (doi: 10.15154/1523041).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.dcn.2023.101334.

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